



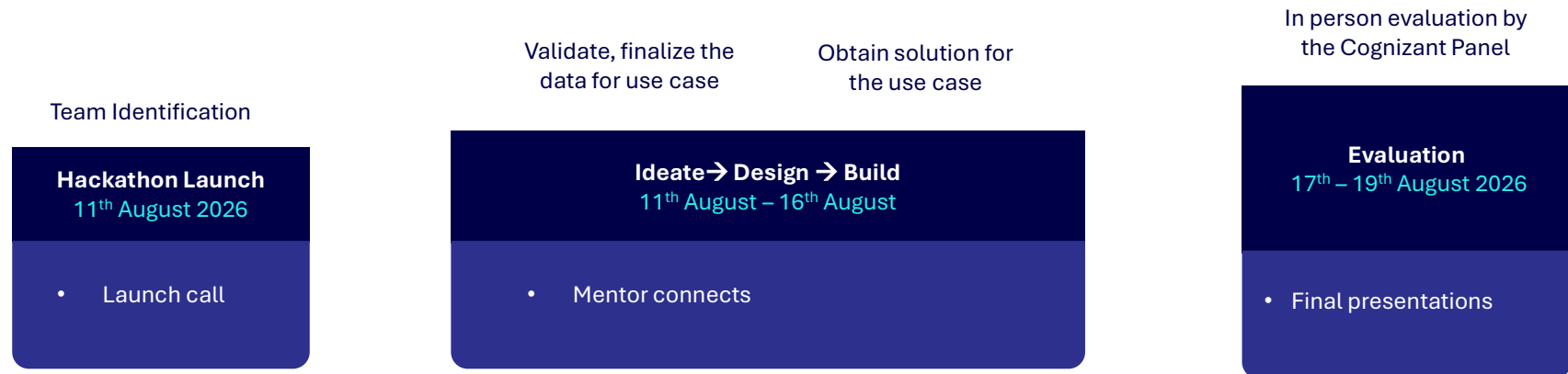
NPN_SCM Hackathon Approach

August 2026



Hackathon Approach

Outcomes



Activities

- Teams with 5-6 members in each team
- 5 use cases are provided; each team should choose 1 combination that contains 2 use cases
- Require one mentor from the college for each of the teams. Cognizant will also assign a mentor for each of the teams

Expectation from Participants

- Connect with mentors and get guidance
- Finalize the approach and data for the use case selected
- Submit the design as per the timelines communicated, in PPT or video form over an email
- Obtain solution as per the timeline



Factors To Consider When You Build the Solution

- ❑ **Use Case Understanding and Relevance** - clarity and depth in terms of understanding the problem statement
- ❑ **Solution Architecture** – quality and feasibility of the overall architecture design, how well do the components interact and integrate with each other
- ❑ **Innovation and Creativity** – level of innovation and creativity in the solution, originality of ideas, innovation .
- ❑ **UI and UX** – usability and design of the UI, ease of navigation, visual appeal etc.
- ❑ **Technical Implementation and Source Code Quality** – code readability, use of best practices, documentation etc.
- ❑ **Model Performance and Evaluation** – model accuracy/precision/recall, F1 score and other relevant metrics
- ❑ **Deployment and Integration** – use of CI/CD pipelines, cloud deployment strategies, integration with APIs etc.
- ❑ **Presentation and Communication** – clarity of communication, organization of the presentation, ability to answer questions and justify design choices
- ❑ **Collaboration and Teamwork** – Team dynamics, distribution of tasks and overall collaboration

Evaluation Criteria - Guideline

Post Build stage

- Use case and process flow understanding
- Proposed architecture and various alternatives
- Breadth of sample data
- Architecture consideration
- Performance
- User experience
- Integration alternatives of the solutions
- Reusable
- Ease of implementation
- Real time decisions
- Process of monitoring
- Way of presenting
- Attendance in various mentor connects

Expectation of Best Solution

Presence of

- ❑ **Architecture, Source code**
- ❑ **User Interface**
- ❑ **Documentation, PPT**
- ❑ **Estimation of development, Roadmap**
- ❑ **Presentation**

Hackathon Use cases

S.No	Use Case	Data Source	Use case Description / Problem Statement	Objective / Outcome	Type
E1 Execution	Smart Restock Inventory Alert System	Simulated inventory transaction data; product master; configurable minimum-stock thresholds. Data can be generated using GenAI with realistic boundary conditions.	Small businesses often rely on manual stock checks, causing missed replenishment opportunities and stockouts. This solution provides a lightweight inventory dashboard that continuously monitors stock levels and triggers alerts whenever an item falls below its configured minimum threshold.	1. Maintain item-wise current inventory and reorder thresholds. WM Solution with Coding 2. Simulate stock consumption, sales, receipts, and adjustments. 3. Compare live stock against the minimum threshold. 4. Trigger email, SMS, or WhatsApp-style alerts for low-stock items. 5. Display a dashboard with inventory status, active alerts, and reorder recommendations. 6. Output: low-stock alert log, item-wise stock view, and reorder action summary.	WM Solution with Coding
E2 Execution	Where's My Truck? – Yard, Dock Door & Delivery Tracker	Simulated WMS feed containing trailer ID, tracking number, current yard location, dock-door availability, appointment/arrival time, load priority, and ETA. Mapping APIs can be used for simulated live truck movement.	Warehouse teams and customers often lack a single real-time view of inbound/outbound truck movement, yard location, trailer availability, and dock-door assignments. This solution combines a customer-facing delivery tracker with an operations dashboard that uses WMS feeds to identify where a truck/trailer is located, estimate arrival time, and recommend the appropriate dock door based on availability and planned arrival.	1. Accept a tracking number, trailer ID, or shipment reference. 2. Display simulated real-time truck location on a map with updated ETA and delivery progress. 3. Receive WMS feed for yard location, trailer status, scheduled arrival, and dock-door availability. 4. Identify the trailer that needs to be docked for each arrival window. 5. Recommend or assign an available dock door based on arrival time, trailer/load type, priority, and dock availability. 6. Alert operations users when a truck is delayed, a dock is unavailable, or reassignment is needed. 7. Output: live map tracker, yard status board, dock-door assignment schedule, ETA alerts, and trailer-to-door allocation summary	WM Solution with Coding

S.no	Use Case	Data Source	Use case Description / Problem Statement	Objective / Outcome	Type/ Key Success Criteria
PR 1 Procurement	Procurement Planning & Supplier Allocation Automation Risk Prediction	Synthetic data of: - Material demand forecast (by SKU/plant) - Approved supplier list - Supplier capacity & lead times - Material prices - Supplier performance scores (OTD, Quality) - Contract commitments (minimum/maximum allocation)	A manufacturing company procures raw materials from multiple suppliers. Demand fluctuations, varying supplier reliability, and cost differences often lead to supply disruptions or excessive procurement costs. Students need to design an optimization model that allocates procurement quantities across suppliers while considering cost, capacity, lead times, quality ratings, and contractual commitments.	To create an optimal sourcing strategy that balances cost, risk, and supplier performance. Expected Outcomes: - Allocate procurement volumes among suppliers efficiently. - Minimize total procurement cost while meeting demand. - Reduce dependency on high-risk suppliers. - Incorporate supplier quality and delivery performance into sourcing decisions. - Provide scenario analysis for supplier disruptions or demand spikes. - Predict potential delivery delays before PO release.	Procurement Optimization Simulation
PR 2 Procurement	End-to-End Autonomous Procure-to-Pay (P2P) System: A fully autonomous, touchless platform designed for high-volume enterprise usage.	Synthetic data of: Intelligent OCR and conversational NLP models. Leverage GenAI engine for data simulation with boundary conditions.	The solution should: Address high-volume bottlenecks using AI capabilities including conversational NLP, intelligent OCR, and predictive anomaly detection. - Requisition: Conversational NLP chatbot for users to request items. - Sourcing & PO: AI selects best supplier and auto-generates PO. - Receiving: Simulated IoT/Computer Vision for goods receipt. - Invoicing: Intelligent OCR and AI for 3-way matching (PO, Receipt, Invoice) to auto-approve payments.	Input: Touchless inputs (user requests, enterprise supplier catalogs, digital invoices). Process: Conversational NLP intent extraction, rule-based PO creation, intelligent OCR data extraction, AI matching. Expected Outcomes: - Touchless invoice processing - Automated Pos - P2P analytics dashboard - Demonstrate successful 3-way matching. Clear presentation of the end-to-end automation.	Primary success criteria: Cost reduction, operational efficiency, and fully autonomous operations.

S.no	Use Case	Data Source	Use case Description / Problem Statement	Objective / Outcome	Type
P1 Planning	Demand Sensing & Replenishment Planning for MedCare Pharma	Synthetic data of – - Historical & sensed demand (by SKU, by region) - Current inventory with batch/expiry details - Warehouse capacity & lead times - Distributor order patterns - Promotional/seasonal calendar	MedCare Pharma faces flu-season demand spikes (+60%) causing stock-outs of critical SKUs in Tier-2 cities, while metro DCs sit on excess near-expiry stock. Siloed regional warehouses and expiry-blind allocation drive both wastage and lost sales. Need: sense near-term demand shifts, allocate stock with expiry awareness, and balance safety stock vs. wastage across DCs.	To build a demand-sensing and replenishment plan for MedCare Pharma that keeps critical SKUs available while minimizing expiry-driven wastage across the network. Expected Outcomes: - Improve short-term forecast accuracy using sensing signals (promotions, seasonality, leading indicators). - Design expiry-aware allocation logic to reduce write-offs. - Optimize replenishment quantity and frequency across DCs. - Reduce stock-outs of critical SKUs during demand surges. - Recommend a review cadence/escalation process for shortage situations	Demand Planning & Replenishment simulation
P2 Planning	Integrated S&OP for TrendWear Apparel	Synthetic data of – - Seasonal demand forecasts by style/SKU - Plant production capacity (units/week) - Fabric lead times & MCQs - DC-to-store transportation/lead-time data - Historical sell-through & markdown data	TrendWear launches a new line every 6 weeks, but forecast misses cause overproduction of slow movers and stock-outs of fast sellers. 4–6-week fabric lead times add rigidity, and disconnected manual S&OP cycles leave Merchandising, Production and Logistics out of sync. Need: align demand, production and procurement plans; time markdowns; and react to in-season sell-through signals.	To design an integrated S&OP process for TrendWear that synchronizes demand, supply and financial plans across the product lifecycle/ Expected Outcomes: - Reconcile merchandising forecasts with plant capacity and fabric lead times. - Build a rolling monthly S&OP cycle integrating sales, production and inventory views. - Recommend markdown timing based on in-season sell-through signals - Optimize fabric procurement against MOQ and lead-time constraints. - Improve cross-functional alignment across Merchandising, Production and Logistics.	S&OP / Integrated Business Planning simulation

Combination of Use Cases:

Each team to take 2 use cases from the below combinations which will be assigned by the BU SMEs

Combination 1: **E1 +P1**

Combination 2: **E2 + PR2**

Combination 3: **E2 + P2**

Combination 4: **P1+PR2**

Combination 5: **P2 + PR1**

E1	Smart Restock Inventory Alert System
E2	Where’s My Truck? – Yard, Dock Door & Delivery Tracker
P1	Demand Sensing & Replenishment Planning for MedCare Pharma
P2	Integrated S&OP for TrendWear Apparel
PR 1	Procurement Planning & Supplier Allocation Automation Risk Prediction
PR 2	End-to-End Autonomous Procure-to-Pay (P2P) System: A fully autonomous, touchless platform designed for high-volume enterprise usage



Thank You